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Polynomization of Sun's conjecture

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Abstract
Let $p(n)$ denote the number of partitions of a natural number n. As $n \to \infty$, the nth root of $p(n)$ tends to 1, which is related to the Cauchy–Hadamard test for power series. Andrews also discovered an elementary proof. Sun conjectured that this happens in a certain way for $n \geq 6$:

$$\sqrt[n]{p(n)} > \sqrt[n+1]{p(n+1)}.$$

The conjecture was proved by Wang and Zhu; shortly thereafter, Chen and Zheng independently obtained a second proof. In this paper, we follow an approach by Rota. We regard $p(n)$ as special values of the D'Arcais polynomials, known as the Nekrasov–Okounkov polynomials. This identifies Sun's conjecture as a property of the largest real zero of certain polynomials. This leads to results toward k-colored partitions, overpartitions, and plane partitions. Moreover, we also consider Chebyshev and Laguerre polynomials. The main purpose of this paper is to offer a uniform approach.

Keywords Inequalities · Partitions · Polynomials · Sequences

Mathematics Subject Classification Primary 05A20 · 11P84; Secondary 05A17 · 11B83

1 Introduction and statement of results

In this paper, we generalize Sun's conjecture [40] on the decreasing monotonicity of the nth root of the partition function $p(n)$. Sun conjectured [40, Conjecture 2.14

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(2012-08-02)] that for all $n \geq 6$ we have

$$\sqrt[n]{p(n)} > \sqrt[n+1]{p(n+1)}.$$

The conjecture has been proven by Wang and Zhu [41] and shortly thereafter by Chen and Zheng [13]. Both proofs depend on the celebrated circle method of Hardy–Ramanujan and the Hardy–Ramanujan–Rademacher formula for $p(n)$ and are closely related to the work by Nicolas [33] and DeSalvo and Pak [17] on the log-concavity of $p(n)$. In this paper, we take a new approach.

In the spirit of Rota [38], we view Sun's conjecture as a claim on the location of zeros of certain associated polynomials.

"The one contribution of mine that I hope will be remembered has consisted in just pointing out that all sorts of problems of combinatorics can be viewed as problems of location of the zeros of certain polynomials and in giving these zeros a combinatorial interpretation. This is now called the critical problem."

The approach provided in this paper is the following. Let g be a normalized arithmetic function with positive values. We assign a sequence of polynomials $\{P_n^g(x)\}_{n\geq 0}$:

$$P_n^g(x) := \frac{x}{n} \sum_{k=1}^{n} g(k) P_{n-k}^g(x) \tag{1.1}$$

with initial value $P_0^g(x) = 1$.

Definition For $n \geq 1$, define

$$\Delta_n^g(x) := \left(P_n^g(x)\right)^{n+1} - \left(P_{n+1}^g(x)\right)^n. \tag{1.2}$$

We denote by x_n^g the largest real zero of $\Delta_n^g(x)$.

The relevance of x_n^g is that it gives a threshold for the positivity of $\Delta_n^g(x)$. Indeed, Lemma 2.2 shows that

$$\Delta_n^g(x) > 0 \quad \text{for all } x > x_n^g.$$

Consequently, an upper bound for x_n^g immediately yields inequalities for special values of the polynomials $P_n^g(x)$. In particular, in the case $g = \sigma$, the inequality $x_n^\sigma < 1$ implies

$$p(n)^{n+1} > p(n+1)^n.$$

1.1 D'Arcais polynomials

The polynomials associated with $g(n) = \sigma(n) = \sum_{d|n} d$ are the D'Arcais polynomials [16, 24]. In combinatorics and statistical mechanics, they are called Nekrasov–Okounkov polynomials [20, 32] and provide a hook length formula. We

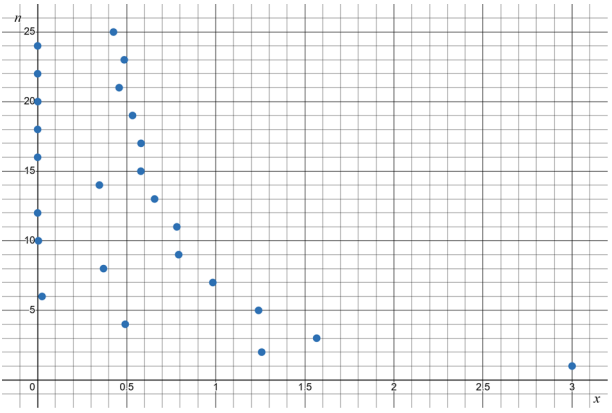


Fig. 1 Largest real zeros of $\Delta_n^\sigma(x)$ for $1 \le n \le 25$

have

$$\sum_{n=0}^{\infty} P_n^\sigma(x)\, t^n = \prod_{n=1}^{\infty} \left(1-t^n\right)^{-x} = \sum_{n=0}^{\infty} \sum_{\lambda \vdash n} t^{|\lambda|} \prod_{h \in H(\lambda)} \left(1 + \frac{x-1}{h^2}\right).$$

Here, $\lambda \vdash n$ denotes that λ is a partition of n and $H(\lambda)$ denotes the multiset of hook lengths associated with $\lambda \vdash n$ and $|\lambda| = n$ is the weight of λ. Therefore, $p(n) = P_n^\sigma(1)$, and more generally $P_n^\sigma(k)$ is the number of k-colored partitions of n. By Lemma 2.2 we have

$$\lim_{x \to \infty} \Delta_n^g(x) = \infty.$$

In Fig. 1, we have plotted the largest real zeros of $\Delta_n^\sigma(x)$ for $1 \le n \le 25$.

We extended our calculations with the support of Steven Charlton and obtain that $x_n^\sigma < 1$ for $6 \le n \le 150$ and speculate that this holds true for all $n \ge 6$. We refer to Sect. 3 for a full picture of our results related to partition numbers and the D'Arcais polynomials.

1.2 Pochhammer and Laguerre polynomials

Already simple input functions $g(n)$ lead to interesting results and give evidence on the significance of studying the largest real zeros x_n^g. Let $\psi_\ell(n) := n^\ell$ for $\ell \in \mathbb{N}_0$. We start with the case $\ell = 0$, which is related to the Pochhammer polynomials.

Theorem 1.1 *Let $g(n) = \psi_0(n)$. Then we obtain*

$$P_n^{\psi_0}(x) = \frac{1}{n!} \prod_{k=0}^{n-1} (x+k) \quad \textit{and } x_n^{\psi_0} = \psi_0(2) = 1 \quad \textit{for } n \ge 2.$$

The proof is given in Sect. 4.1. This result implies that $P_n^{\psi_0}(x)^{n+1} > P_{n+1}^{\psi_0}(x)^n$ for all $x > 1$ and $n \ge 2$. More involved is the next case, $g(n) = \psi_1(n)$, which is related to

orthogonal polynomials, the α-associated Laguerre polynomials for $\alpha = 1$:

$$L_n^{(\alpha)}(x) := \sum_{k=0}^{n} \binom{n+\alpha}{n-k} \frac{(-x)^k}{k!}.$$

From [23], we have $P_n^{\psi_1}(x) = \frac{x}{n} L_{n-1}^{(1)}(-x)$. Utilizing a Turán inequality of $P_n^{\psi_1}(x)$ [26] and an initial property of $\Delta_n^{\psi_1}(x)$ we will prove in Sect. 4.2 the following result.

Theorem 1.2 *Let $g(n) = \psi_1(n)$. Then $P_n^{\psi_1}(x) = \frac{x}{n} L_{n-1}^{(1)}(-x)$ for $n \geq 1$. The largest real zero of $\Delta_n^{\psi_1}(x)$ satisfies $x_n^{\psi_1} \leq \psi_1(2) = 2$ for all $n \in \mathbb{N}$. Further, for all $n \geq 6$, we have $x_n^{\psi_1} < 1$.*

This implies that for all $x \geq 1$ and $n \geq 6$:

$$\frac{x^{n+1}}{n^{n+1}} L_{n-1}^{(1)}(-x)^{n+1} > \frac{x^n}{(n+1)^n} L_n^{(1)}(-x)^n.$$

It would be interesting to investigate this new type of inequality for other orthogonal polynomials, such as Chebyshev and Hermite polynomials.

1.3 Plane partitions and overpartitions

1.3.1 Plane partitions

Recently, Ono et al. [35] proved a conjecture [27] on the log-concavity of the sequence of plane partition numbers $\{pp(n)\}_n$ for $n \geq 12$. This strongly suggests to get similar results for associated polynomials $P_n^{\sigma_2}(x)$ as we get for D'Arcais polynomials $P_n^{\sigma}(x)$. At present, however, we have only partial results, although our numerical data draw a clear picture. Plane partitions are fascinating generalizations of integer partitions [39, Sect. 7.20]. Andrews [5] also introduced plane partitions in the context of higher-dimensional partitions.

We follow the introduction on plane partitions provided in [27]. A plane partition π of n is an array $\pi = \left(\pi_{ij}\right)_{i,j\geq 1}$ of non-negative integers π_{ij}, with finite sum $|\pi| := \sum_{i,j\geq 1} \pi_{ij} = n$, which is weakly decreasing in rows and columns. It can be considered as the filling of a Ferrers diagram with weakly decreasing rows and columns, where the sum of all these numbers equals n. Let the numbers in the filling represent the heights for stacks of blocks placed at the corresponding cell of the diagram (Fig. 2). We refer to [27]. This is a natural generalization of the concept of classical partitions [5, 34].

Let $pp\,(n)$ denote the number of plane partitions of n. Similarly as Euler discovered the generating function for the partition numbers $p(n)$, MacMahon [31] discovered the generating function for the plane partitions $pp\,(n)$:

$$\prod_{n=1}^{\infty} \left(1-q^n\right)^{-n} = \sum_{n=0}^{\infty} pp\,(n)\; q^n.$$

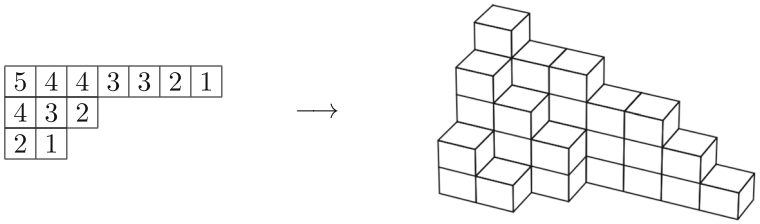


Fig. 2 Representation of plane partitions

Table 1 Approximate values of $\sqrt[n]{pp\,(n)}$

n	1	2	3	4	5	6	7	8	9	10
$\sqrt[n]{pp\,(n)}$	1	1.732	1.817	1.899	1.888	1.906	1.890	1.886	1.872	1.862

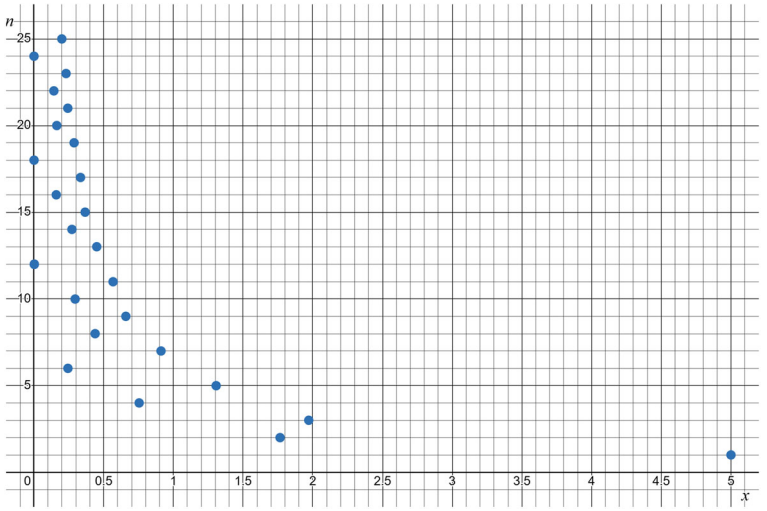


Fig. 3 Largest real zeros of $\left(P_n^{\sigma_2}(x)\right)^{n+1} - \left(P_{n+1}^{\sigma_2}(x)\right)^n$

Therefore, let $g(n) = \sigma_2(n)$, where $\sigma_\ell(n) := \sum_{d|n} d^\ell$. Then we have $pp\,(n) = P_n^{\sigma_2}(1)$. We list the first values of $\sqrt[n]{pp\,(n)}$ in Table 1.

Further, we have plotted the largest real zeros of $\Delta_n^{\sigma_2}(x)$ for $1 \le n \le 25$ (see Fig. 3). Then this suggests:

Conjecture 1 *Let $g(n)$ be equal to $\sigma_2(n)$. Then we have that $\Delta_n^{\sigma_2}(x) > 0$ for $x > g\,(2) = 5$ and all $n \in \mathbb{N}$. Moreover, for $n \ge 6$, we already have $\Delta_n^{\sigma_2}(x) > 0$ for $x \ge 1$.*

We prove the conjecture for $x = 1$ utilizing the log-concavity of $pp\,(n)$ for $n \ge 12$ by a similar argument as given in [41].

Theorem 1.3 *The sequence $\left\{\sqrt[n]{pp\,(n)}\right\}_n$ is strictly decreasing for $n \ge 6$.*

Table 2 Values $P_n^{\bar{g}}(x)$

n	1	2	3	4	5	6	7
$P_n^{\bar{g}}(-2)$	-2	0	0	2	0	0	0
$P_n^{\bar{g}}(-1)$	-1	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{3}{8}$	$\frac{1}{8}$	$\frac{3}{16}$	$\frac{7}{16}$
$P_n^{\bar{g}}(1)$	1	$\frac{3}{2}$	$\frac{5}{2}$	$\frac{27}{8}$	$\frac{39}{8}$	$\frac{111}{16}$	$\frac{149}{16}$
$P_n^{\bar{g}}(2)$	2	4	8	14	24	40	64

1.3.2 Overpartitions

In our investigation of overpartitions, we have observed analogous patterns. An overpartition of n is a non-increasing sequence of natural numbers that sum to n where the last occurrence of a number may be overlined [15]. We denote the number of overpartitions of n by $\bar{p}(n)$. The following generating function is well-known:

$$\sum_{n=0}^{\infty} \bar{p}(n)\, t^n = \prod_{n=1}^{\infty} \frac{1+t^n}{1-t^n} = 1 + 2t + 4t^2 + 8t^3 + 14t^4 + \cdots$$

We offer the following polynomization

$$\sum_{n=0}^{\infty} P_n^{\bar{g}}(x)\, t^n = \left(\frac{\prod_{n=1}^{\infty}(1+t^n)}{\prod_{n=1}^{\infty}(1-t^n)} \right)^{\frac{x}{2}} = \left(\sum_{n=-\infty}^{\infty} (-1)^n\, t^{n^2} \right)^{-\frac{x}{2}}.$$

In Table 2, we provide several values of $P_n^{\bar{g}}(x)$. Note that $\bar{p}(n) = P_n^{\bar{g}}(2)$ denotes the number of overpartitions of n and more generally $P_n^{\bar{g}}(2k)$ corresponds to the number of k-colored overpartitions of n for $k \in \mathbb{N}$.

The distribution of the largest zeros $x_n^{\bar{g}}$ (see Fig. 4) suggests that $x_n^{\bar{g}} < 1$ for $n \geq 6$. For $x = 2$, we prove:

Theorem 1.4 *We have that* $\Delta_1^{\bar{g}}(2) = \Delta_2^{\bar{g}}(2) = 0$. *Let* $n \geq 3$. *Then*

$$\bar{p}(n)^{n+1} > (\bar{p}\,(n+1))^n\,.$$

Therefore, $\Delta_n^{\bar{g}}(2) > 0$ *for* $n \geq 3$.

2 Preliminaries

2.1 Basic properties of sequences

Let $\{\alpha(n)\}_{n=0}^{\infty}$ be a sequence of positive real numbers. We consider roots and quotients of sequence members. This has already been a successful approach in analysis finding

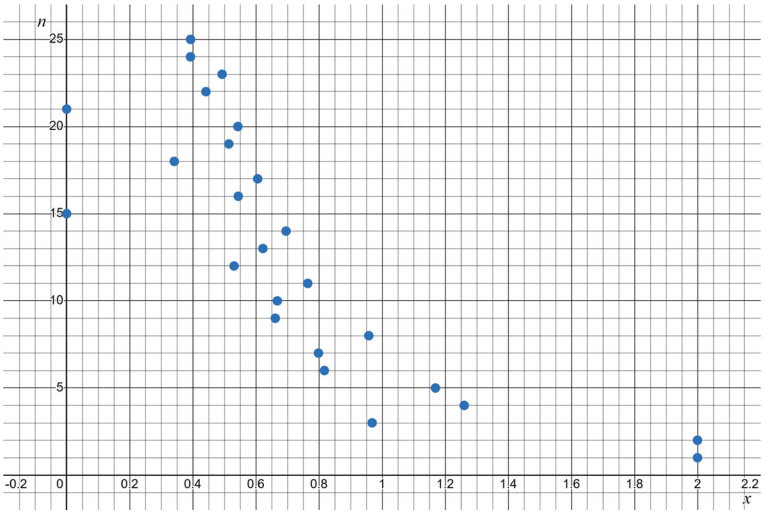


Fig. 4 Largest real zeros of $\Delta_n^{\bar{g}}(x)$ for $1 \le n \le 25$

the radius of convergence in the associated power series $\sum_{n=0}^{\infty} \alpha(n)\, t^n$:

$$R_\alpha(n) := \sqrt[n]{\alpha(n)} \quad \text{and} \quad Q_\alpha(n) := \frac{\alpha(n)}{\alpha(n-1)}.$$

Then we obtain for $n \ge 2$:

$$R_\alpha\,(n) > Q_\alpha\,(n) \Leftrightarrow R_\alpha\,(n-1) > R_\alpha\,(n) \Leftrightarrow R_\alpha\,(n-1) > Q_\alpha\,(n)\,. \tag{2.1}$$

A sequence is called log-concave for $n \ge n_0$ if

$$\alpha(n)^2 - \alpha(n-1)\alpha(n+1) \ge 0$$

for all $n \ge n_0$. It is called strictly log-concave if the inequality is strict.

Proposition 2.1 *Let the sequence $\{\alpha(n)\}_n$ of positive real numbers be log-concave for all $n \ge n_0$ for $n_0 \in \mathbb{N}$, $n_0 \ge 2$. Then, the initial condition*

$$R_\alpha\,(n_0 - 1) > R_\alpha\,(n_0) \tag{2.2}$$

implies that the sequence $\{\sqrt[n]{\alpha(n)}\}_{n=n_0-1}^{\infty}$ is strictly decreasing.

Proof A sequence $\{\alpha(n)\}_n$ is log-concave at n if and only if $Q_\alpha(n) \ge Q_\alpha(n+1)$. From the general property (2.1), we observe that the initial condition is equal to $R_\alpha(n_0) > Q_\alpha(n_0)$. And applying again (2.1) leads to $R_\alpha(n_0) > R_\alpha(n_0+1)$. The proof is completed by applying mathematical induction. □

If we consider $1/\alpha(n)$, we obtain similar results for log-convex sequences. Wang and Zhu first observed that the monotonicity of quotients implies the monotonicity of the roots stated in [41, Theorem 2.1]. Note that Proposition 2.1 implies the result by Wang and Zhu.

2.2 Basic properties of generalized D'Arcais polynomials

Let g be a normalized arithmetic function with positive values. We have introduced in the introduction [see Definition (1.1)], associated with g, the generalized D'Arcais polynomials $\{P_n^g(x)\}_{n\geq 0}$. Then, the location of the largest real zero x_n^g of

$$\Delta_n^g(x) = \left(P_n^g(x)\right)^{n+1} - \left(P_{n+1}^g(x)\right)^n,$$

determines the positivity of $\Delta_n^g(x)$ since:

Lemma 2.2 *Let the sequence $\{P_n^g(x)\}_n$ be given. Let x_n^g be the largest real zero of $\Delta_n^g(x)$. Then $\Delta_n^g(x) > 0$ for all $x > x_n^g$.*

Proof Let $n \geq 1$. Note that real zeros exist since $P_n^g(0) = 0$. The degree of $P_n^g(x)$ is equal to n and is determined by $P_n^g(x) = \frac{1}{n!}\sum_{k=1}^n A_{n,k}^g\, x^k$ with $A_{n,n}^g = 1$. Consequently, the leading coefficient of $\Delta_n^g(x)$ is given by:

$$\left(\frac{1}{n!}\,A_{n,n}^g\right)^{n+1} - \left(\frac{1}{(n+1)!}A_{n+1,n+1}^g\right)^n = \left(\frac{1}{n!}\right)^{n+1} - \left(\frac{1}{(n+1)!}\right)^n > 0.$$

□

In this paper, we provide evidence that $x_n^g \leq g(2)$ for polynomials related to partition numbers, plane partition numbers, overpartitions including Pochhammer polynomials and associated Laguerre polynomials. We provide proofs in some cases but are far from the general case. We believe that new techniques have to be discovered. Nevertheless, for $n = 1$ and $n = 2$, we are explicit and obtain some necessary conditions.

Remark We have that

$$\begin{aligned}
\Delta_1^g(x) &= \frac{x}{2}\,(x - g(2))\,,\\
\Delta_2^g(x) &= \frac{x^2}{72}\left(7x^4 + 15g(2)\,x^3 + \left(9\,(g(2))^2 - 8g(3)\right)x^2\right.\\
&\qquad \left. + \left(9\,(g(2))^3 - 24g(3)\,g(2)\right)x - 8\,(g(3))^2\right).
\end{aligned}$$

Lemma 2.3 *We have that $\Delta_2^g(x) \geq 0$ for all $x \geq g(2)$ if and only if $g(3) \leq (g(2))^2$.*

Proof Let $\Delta_2^g(x) = \frac{x^2}{72}Q(x)$. Then $Q(x + g(2)) = 7x^4 + 43g(2)\,x^3 + (96\,(g(2))^2 - 8g(3))\,x^2 + \left(100\,(g(2))^3 - 40g(3)\,g(2)\right)x + 40\,(g(2))^4 - 32g(3)\,(g(2))^2 -$

$8\,(g\,(3))^2$. Suppose $g\,(3) \le (g\,(2))^2$. Then $96\,(g\,(2))^2 - 8\,g\,(3) \ge 88\,(g\,(2))^2$ and $100\,(g\,(2))^3 - 40\,g\,(3)\,g\,(2) \ge 60\,(g\,(2))^3$. As

$$40\,(g\,(2))^4 - 32 g\,(3)\,(g\,(2))^2 - 8\,(g\,(3))^2 = 8\left((g\,(2))^2 - g\,(3)\right)\left(5\,(g\,(2))^2 + g\,(3)\right) \tag{2.3}$$

we have

$$Q\,(x) \ge 7x^4 + 43\,g\,(2)\,x^3 + 88\,(g\,(2))^2\,x^2 + 60\,(g\,(2))^3\,x \ge 0 \quad \text{for } x \ge 0.$$

If $g\,(3) > (g\,(2))^2$ then $Q\,(0) < 0$ by (2.3). □

The arithmetic functions $\sigma_\ell(n) := \sum_{d|n} d^\ell$ and especially the case $\ell = 1$ lead to Sun's conjecture. In this case, we recover the D'Arcais polynomials [16, 24], which are related to the q-expansion of powers of the Dedekind function. These polynomials are also known as the Nekrasov–Okounkov polynomials [20, 32] in statistical mechanics and combinatorics. Note that $P_n^{\sigma_1}\,(1) = p(n)$ and $P_n^{\sigma_2}\,(1) = pp(n)$, where $pp(n)$ is the number of plane partitions of n. We refer to [5]. Several other interesting cases indicate a general pattern of $\Delta_n^g(x)$ (1.2) regarding the location of the largest real zero of polynomials defined by (1.1). The number of ℓ-tuples of permutations in S_n, which generate abelian subgroups, is related to $g_\ell(n) := \sum_{d|n} d g_{\ell-1}(d)$. Here $g_0(n) = 1$ for $n = 1$ and 0 otherwise [1, 2, 12, 28]. Moreover, let $\bar{g}(n) := \sigma_1(n) - \sigma_1(n/2)$ with $\sigma_1\,(x) = 0$ for $x \notin \mathbb{N}$. Then $P_n^{\bar{g}}(2) = \bar{p}(n)$ [15] is the number of overpartitions of n.

3 Partitions and D'Arcais polynomials

Let $p(n)$ denote the number of partitions of n. A partition λ of a positive integer n is any non-increasing sequence $(\lambda_1, \lambda_2, \ldots, \lambda_d)$ of positive integers whose sum $|\lambda| := \sum_{k=1}^{d} \lambda_k$ is equal to n. We refer to the works by Andrews and Ono [5, 34]. Euler's generating function

$$\sum_{n=0}^{\infty} p(n)\,q^n = \prod_{n=1}^{\infty} \frac{1}{1-q^n} = 1 + q + 2q^2 + 3q^3 + 5q^4 + \cdots$$

represents a holomorphic function in the unit disk [22], [7, Sect. 14.3], [37, Chap. 1]. By Cauchy–Hadamard's theorem, the sequence of partition numbers $\{p(n)\}$ [6, Sect. 6.4] exhibits a subexponential function, satisfying

$$\lim_{n\to\infty} \sqrt[n]{p(n)} = 1. \tag{3.1}$$

Andrews demonstrated that this property can also be derived through combinatorial methods [4, 6]. The first values of $p(n)$ (Table 3) show some irregularity.

Sun [40] observed and conjectured that the sequence $\sqrt[n]{p(n)}$ satisfies the following property which was proven by Wang and Zhu [41]; shortly thereafter, Chen and Zheng [13] obtained an independent proof.

Table 3 Approximate values of $\sqrt[n]{p(n)}$

n	1	2	3	4	5	6	7
$\sqrt[n]{p(n)}$	1.00	1.41	1.44	1.50	1.48	1.49	1.47

Theorem 3.1 (Wang and Zhu [41]) *The sequence* $\{\sqrt[n]{p(n)}\}_n$ *is strictly decreasing for* $n \geq 6$.

Proof We consider the sequence $\sqrt[n]{p(n)}_{n\geq 1}$. Nicolas [33] (see also DeSalvo and Pak [17]) proved the log-concavity of $\{p(n)\}$ for $n \geq 26$. Therefore, we can apply Proposition 2.1 for $n_0 = 26$. The remaining cases $6 \leq n \leq 24$ can be checked directly by a computer. □

This proof method we call the *log-concavity method with initial condition*, discovered by Wang and Zhu.

Remark The only known proof of the log-concavity of the sequence $\{p(n)\}_n$ for $n \geq 26$ relies on the celebrated circle method by Hardy and Ramanujan [21], utilizing Rademacher's explicit formula [36].

The strictly decreasing property also reads as

$$p(n)^{n+1} > p(n+1)^n, \tag{3.2}$$

which may have some combinatorial interpretation. Encouraged by Andrews' combinatorial proof [6, Sect. 6.4] of (3.1), we believe it is worth examining whether inequality (3.2) also admits a combinatorial proof. Notably, combinatorial proofs had been provided for the Bessenrodt–Ono inequality [8] by Alanazi et al. [3] as well as by Chern et al. [14]. If one considers the partition numbers $p(n)$ as a special value of the nth D'Arcais polynomial $P_n^\sigma(x)$, where $p(n) = P_n^\sigma(1)$, it is natural to investigate the distribution of the zeros of $\Delta_n^\sigma(x)$. Since $\lim_{x\to\infty} \Delta_n^\sigma(x) = \infty$, it is obvious that $\Delta_n^\sigma(x) > 0$ for $x > x_n^\sigma$, where x_n^σ is the largest real zero of $\Delta_n^\sigma(x)$. We refer to Sect. 1.1 for a plot of the largest real zero for $1 \leq n \leq 25$. More generally, we have with the support of Steven Charlton:

Theorem 3.2 *Let* $6 \leq n \leq 150$. *Then* $x_n^\sigma < 1$.

In PARI/GP the following code generates the results:

```
M=150; p=vector(M+2); p[1]=1;
for(n=1,M+1,p[n+1]=(x/n)*sum(k=1,n,sigma(k)*p[n-k+1]));
Delta=vector(M,n,p[n+1]^(n+1)-p[n+2]^n);
xsigma=vector(M,n,vecmax(polrootsreal(Delta[n]))<1);
```

Then $x_n^\sigma < 1$ if `xsigma[n]` $= 1$.
Further calculations lead to the following observation.

Conjecture 2 *Let* $n \geq 6$ *and* $x \geq 1$. *Then*

$$(P_n^\sigma(x))^{n+1} - (P_{n+1}^\sigma(x))^n > 0.$$

Let $k \in \mathbb{N}$. We denote by $p_k(n)$ the number of k-colored partitions of n [9, 10, 14]. A k-colored partition of n is a partition of n where to each part one of k colors is allocated. Note that $p_1(n) = p(n)$. The D'Arcais polynomials evaluated at k generalize Euler's original approach toward the number of partitions $p(n)$. We have $P_n^{\sigma}(k) = p_k(n)$. Therefore, Theorem 3.2 implies that $\{\sqrt[n]{p_k(n)}\}_n$ is strictly decreasing for $6 \leq n \leq 150$ for all k-colored partitions. We prove Conjecture 2 for k-colored partitions. We utilize a deep result by Bringmann et al. [10] confirming the Chern–Fu–Tang conjecture [14]. We refer also to [25, (1.2)].

Theorem 3.3 *Let $n, k \in \mathbb{N}$ and $n \geq 6$. Then $\Delta_n^{\sigma}(k) > 0$. Therefore, for all k-colored partitions and all $n \geq 6$:*

$$p_k(n)^{n+1} > p_k(n+1)^n. \tag{3.3}$$

Proof The 1-colored partitions are considered in Theorem 3.1. Therefore, let $k \geq 2$. Then, [10] implies that $p_2(n)$ is log-concave for $n \geq 6$ and $p_k(n)$ is log-concave for $k \geq 3$ and all n. Since $x_5^{\sigma} < 2$, we can apply Lemma 2.2 and conclude the proof. □

Corollary 3.4 *Let $k \geq 2$. Then, the strict inequality (3.3) holds true for all natural numbers n except for $k = 2$ and $n = 1$ holds the opposite inequality and for $k = 3$ and $n = 1$ holds equality.*

Proof For $n = 1$, we can check the cases $k = 2, 3$ directly, and when $k \geq 4$, we apply Lemma 2.2 with $x_1^{\sigma} = 3$. For $n = 2, 3, 4$, a direct computation gives $x_2^{\sigma}, x_3^{\sigma}, x_4^{\sigma} < 2$; these values are displayed in Fig. 1. □

Remarks

(i) Let $k \geq 2$. Then, the sequence $\left\{\sqrt[n]{p_k(n)}\right\}_{n \geq 1}$ is strictly decreasing except in the cases $k \in \{2, 3\}$ only for $n \geq 2$.
(ii) Note, it is not necessary to have strict log-concavity [9]. Moreover, the inequality $p_2(5)^2 < p_2(4)\, p_2(6)$ has no negative effect.
(iii) The general result of [10] on the log-concavity of $P_n^{\sigma}(x)$ is not sufficient to prove a result for $x \geq 2$ and $x \notin \mathbb{N}$, since one also needs the initial condition related to Proposition 2.1 following the proof method of [41].
(iv) A direct proof of the claim $x_n^{\sigma} < 2$ for $2 \leq n \leq 5$ would give a new proof of Theorem 3.3 and the Conjecture.

4 The cases $\psi_\ell(n)$: Pochhammer and Laguerre polynomials

For $\ell = 0$, the most explicit case, we have a closed formula for the polynomials $P_n^{\psi_0}(x)$ obtained from the defining recursion (1.1). These are the Pochhammer polynomials. This makes it possible to determine $x_n^{\psi_0}$. Associated Laguerre polynomials are obtained for $\ell = 1$. To determine the location of $x_n^{\psi_1}$ is already a non-trivial task. It is an open question how to attack the cases $\ell > 1$.

4.1 Largest real zero of $\Delta_n^{\psi_0}(x)$

Proposition 4.1 *Let $g(n) = \psi_0(n) = 1$ for all n. Then the largest zero of*

$$\Delta_n^{\psi_0}(x) = \left(P_n^{\psi_0}(x)\right)^{n+1} - \left(P_{n+1}^{\psi_0}(x)\right)^n$$

is $x_n^{\psi_0} = 1$ for all $n \geq 1$.

Proof We obtain

$$\begin{aligned}\Delta_n^{\psi_0}(x) &= \left(\prod_{k=0}^{n-1}\frac{x+k}{k+1}\right)^{n+1} - \left(\prod_{k=0}^{n}\frac{x+k}{k+1}\right)^n \\ &= \left(\prod_{k=0}^{n-1}\frac{x+k}{k+1}\right)^n \left(\prod_{k=0}^{n-1}\frac{x+k}{k+1} - \left(\frac{x+n}{n+1}\right)^n\right).\end{aligned}$$

Therefore, its zeros are $0, -1, \ldots, 1-n$, each n-fold and the zeros of

$$\Pi_n(x) = \prod_{k=0}^{n-1}\left(\frac{x-1}{k+1}+1\right) - \left(\frac{x-1}{n+1}+1\right)^n.$$

Note that it has a zero at $x = 1$ since both terms in the difference are 1 in this case. If $x > 1$ and $k < n$, then $\frac{x-1}{k+1} > \frac{x-1}{n+1}$. Thus, $\Pi_n(x) > 0$, confirming that $x = 1$ is the largest zero in this case. □

4.2 Largest real zero of $\Delta_n^{\psi_1}(x)$

Let $g(n) = \psi_1(n)$. Then, the associated polynomials $P_n^{\psi_1}(x)$ are related to 1-associated Laguerre polynomials $L_{n-1}^{(1)}(x)$. We refer to Sect. 1.2 and [26].

Proof of Theorem 1.2 The polynomials $P_n^{\psi_1}(x)$ satisfy Turán inequalities [26, Theorem 1.4] for all non-negative real numbers x and $n \geq 2$:

$$P_n^{\psi_1}(x)^2 \geq P_{n-1}^{\psi_1}(x)\, P_{n+1}^{\psi_1}(x).$$

Nevertheless, the initial condition (2.2) has some *delay*. It is sufficient to consider the plot Fig. 5 with $1 \leq n \leq 25$. We have $x_1^{\psi_1} = \psi_1(2) = 2$ and $x_n^{\psi_1} < 2$ otherwise. Therefore, we can conclude the result for all $x \geq 2$ for $n \geq 2$. Further, for all $n \geq 6$, we have $x_n^{\psi_1} < 1$.

□

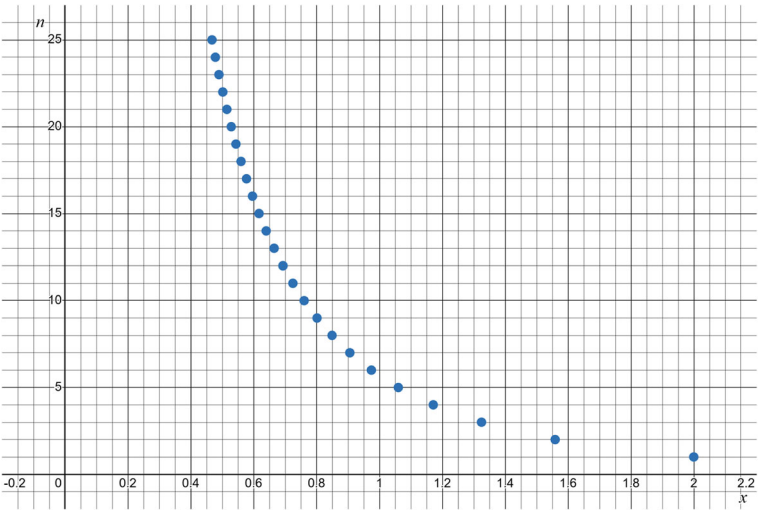


Fig. 5 Largest real zeros of $\Delta_n^{\psi_1}(x)$ for $1 \le n \le 25$

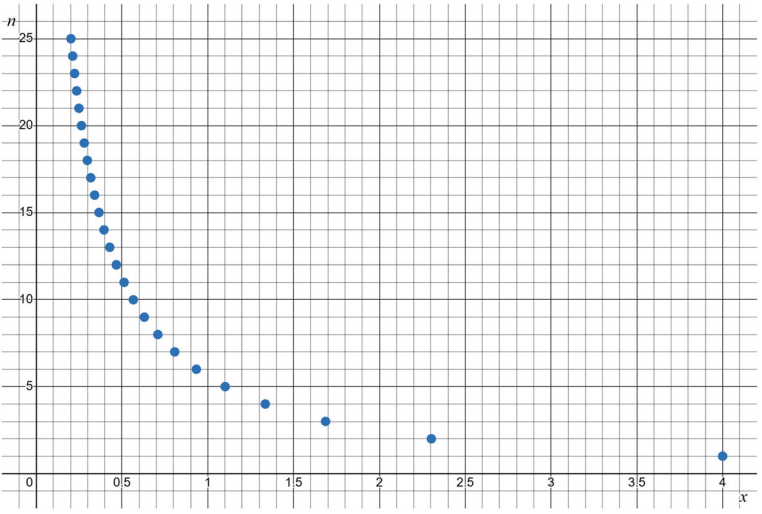


Fig. 6 Largest real zeros of $\Delta_n^{\psi_2}(x)$ for $1 \le n \le 25$

4.3 Largest real zero of $\Delta_n^{\psi_2}(x)$

We add the case $g(n) = \psi_2(n) = n^2$, which shows the same pattern as for $\psi_1(n)$ (Fig. 6).

5 Plane partitions and overpartitions

5.1 Proof of Theorem 1.3

Proof We know already by Ono et al. [35] that $\{pp(n)\}_n$ is log-concave for $n \geq 12$. This has been previously conjectured by Tröger and the two authors of this paper [27, Sect. 1.2]. Since $x_n^{\sigma_2}$ is positive and smaller than 1 for $6 \leq n \leq 11$ (see Fig. 3) we can apply Proposition 2.1 and complete the proof. □

The computations displayed in Fig. 3 indicate that

$$P_n^{\sigma_2}(x)^{n+1} > P_{n+1}^{\sigma_2}(x)^n \tag{5.1}$$

for $x \geq 1$ and all $6 \leq n \leq 25$. Therefore, Ref. [27, Conjecture 2 in Sect. 1.3.2] would imply that the polynomization (5.1) of Theorem 1.3 is also valid for all $n \geq 6$ and $x \geq 1$.

5.2 Proof of Theorem 1.4

Proof The overpartition function $\bar{p}(n)$ is log-concave for all $n \in \mathbb{N}$. Note that $\bar{p}(0) = 1$, $\bar{p}(1) = 2$ and $\bar{p}(2) = 4$.

Theorem 5.1 (Engel [19]) *The function $\bar{p}(n)$ is log-concave for $n \geq 1$.*

We have that $x_1^{\bar{g}} = x_2^{\bar{g}} = 2$ and $x_n^{\bar{g}} < 2$ for $3 \leq n \leq 25$. Again, applying Proposition 2.1 proves Theorem 1.4. □

6 Commuting ℓ-tuples of permutations

We consider the double sequence $\{N_\ell(n)\}$ of positive integers provided by properties of the symmetric group S_n acting on n elements and its abelian subgroups, as measured by its generators. We refer to [1, 2]. Let $\ell, n \geq 1$ and

$$C_{\ell,n} := \left\{ \pi = (\pi_1, \ldots, \pi_\ell) \in S_n^\ell \,:\, \pi_j \pi_k = \pi_k \pi_j \text{ for } 1 \leq j, k \leq \ell \right\}.$$

Each $\pi \in C_{\ell,n}$ corresponds to a group homomorphism from $\mathbb{Z}^\ell$ to S_n. We then define

$$N_\ell(n) := \frac{|C_{\ell,n}|}{n!} = \frac{\left|\text{Hom}\left(\mathbb{Z}^\ell, S_n\right)\right|}{n!} \in \mathbb{N}, \quad \text{with } N_\ell(0) := 1.$$

Dey [18] and Wohlfahrt [42] offer a compelling recursive approach determining the number of homomorphisms $n!\, N_G(n) := |\text{Hom}\,(G, S_n)|$ of a finitely generated group G into S_n. Let $g_G(n)$ denote the number of subgroups of G with index n. Notably, there has been extensive research by Lubotzky and Segal [30] studying the growth of

Table 4 Values for $0 \le n \le 10$

n	0	1	2	3	4	5	6	7	8	9	10
$N_2(n)$	1	1	2	3	5	7	11	15	22	30	42
$N_3(n)$	1	1	4	8	21	39	92	170	360	667	1316
$N_4(n)$	1	1	8	21	84	206	717	1810	5462	13,859	38,497

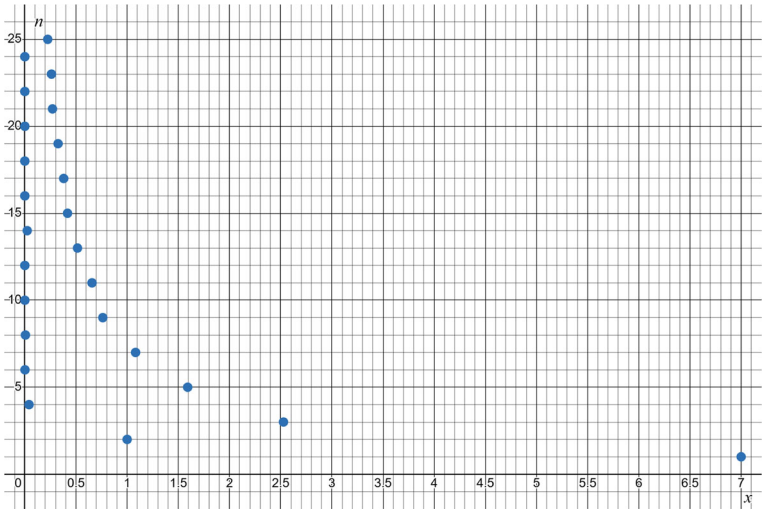


Fig. 7 Largest real zeros of $\Delta_n^{g_3}(x) = \left(P_n^{g_3}(x)\right)^{n+1} - \left(P_{n+1}^{g_3}(x)\right)^n$

these numbers in the context of characterizing the group G itself:

$$N_G(n) = \frac{1}{n} \sum_{k=1}^{n} g_G(k)\, N_G(n-k), \tag{6.1}$$

with the initial value $N_G(0) := 1$. Let $G = \mathbb{Z}^\ell$ and $g_\ell(n) := g_G(n)$. Then $g_1(n) = 1$ and $g_2(n) = \sigma(n) = \sum_{d|n} d$. Therefore, we recover in the case $\ell = 2$ from (6.1) Euler's well known recursion formula for the partition numbers $p(n)$ [5, 34]. The case $\ell = 3$ reveals a connection to topology, as highlighted by Britnell (see the introduction of [11]) due to work by Liskovets and Mednykh [29]. They discovered that $N_3(n)$ counts the number of non-equivalent n-sheeted coverings of a torus. Additionally, the numbers $N_\ell(n)$ were identified by Bryan and Fulman [12] as the nth orbifold characteristics. This generalizes the work by Hirzebruch and Höfer [28] concerning the ordinary and string-theoretic Euler characteristics of symmetric products.

To illustrate the growth of $\{N_\ell(n)\}_{\ell,n}$, we present the first values for $0 \le n \le 10$ and $2 \le \ell \le 4$ in Table 4. Note that $N_1(n) = 1$.

Let $g_\ell(n)$ be the number of subgroups of $\mathbb{Z}^\ell$ of index n. In this subsection, we examine the cases $\ell = 3$ and $\ell = 4$. See Figs. 7 and 8.

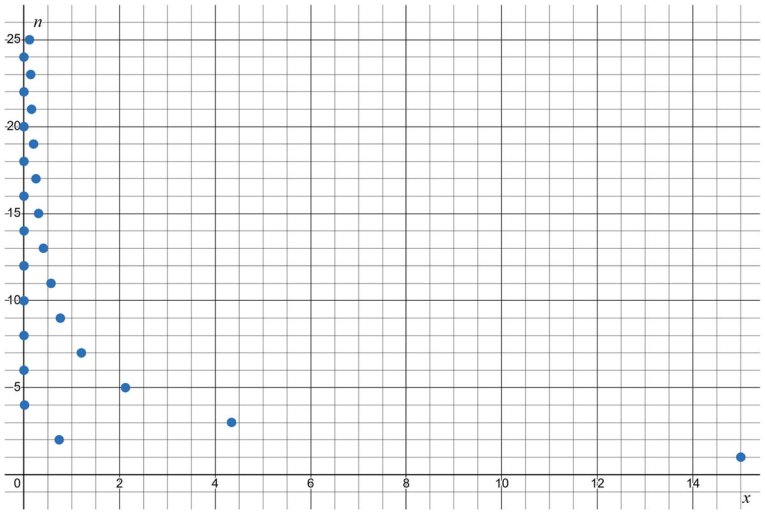


Fig. 8 Largest real zeros of $\Delta_n^{g4}(x) = \left(P_n^{g4}(x)\right)^{n+1} - \left(P_{n+1}^{g4}(x)\right)^n$

Theorem 6.1 *Let* $8 \le n \le 25$. *Then* $\Delta_n^{g3}(x) > 0$ *for all* $x \ge 1$. *Let* $8 \le n \le 25$. *Then* $\Delta_n^{g4}(x) > 0$ *for all* $x \ge 1$.

Ultimately, we state the following speculation which appears to be evident from Figs. 7 and 8.

Conjecture 3 *Let* $a_3 = 8$ *and* $a_4 = 8$. *Let* $\ell = 3$ *or* $\ell = 4$. *Then* $\Delta_n^{g\ell}(x) > 0$ *for all* $x \ge 1$ *for* $n \ge a_\ell$.

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